Project Initialization and Planning Phase

Date	30 June 2024
Team ID	xxxxxx
Project Title	A Comprehensive Measure of Well-Being
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to predict the **Human Development Index (HDI)** using Machine Learning techniques, improving the speed and accuracy of HDI estimation. The system analyzes important socioeconomic indicators and provides quick, reliable predictions through a Flask web application. It helps users reduce manual calculations and supports better planning and decision-making.

Project Overview	
Objective	The primary objective is to develop a Machine Learning-based web application that accurately predicts the Human Development Index (HDI) using key development indicators and provides instant prediction results.
Scope	The project focuses on analyzing HDI-related data, training a Machine Learning model, and deploying it through a Flask web application to provide fast and accurate HDI predictions for users.
Problem Statement	
Description	Manual calculation and analysis of the Human Development Index require multiple socio-economic indicators and significant time. Existing methods are less efficient for quick prediction and comparison.

Impact	Developing an automated HDI prediction system improves analysis speed, reduces manual effort, minimizes calculation errors, and supports researchers, government organizations, and policy makers in making informed decisions.
Proposed Solution	
Approach	Develop a Machine Learning model that predicts the Human Development Index based on Life Expectancy, Mean Years of Schooling, Expected Years of Schooling, and Gross National Income (GNI). The trained model is integrated with a Flask web application to provide real-time predictions.
Key Features	• Machine Learning-based HDI prediction. • User-friendly Flask web interface. • Real-time prediction of HDI score. • Automatic classification into Low, Medium, High, or Very High Human Development. • Fast, accurate, and reliable prediction results.

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU	Intel Core i5 / Ryzen 5 or above
Memory	RAM	8 GB
Storage	Disk Space	500 GB SSD
Software		
Frameworks	Web frameworks	Flask
Libraries	ML libraries	scikit-learn, NumPy, Pandas, Pickle
Development Environment	IDE	Visual Studio Code / PyCharm / Jupyter Notebook
Data		

Data	Data Source, size, format	Human Development Index Dataset, CSV, 195 records, 5 attributes